Blurt sheet - Unit 3.4 Operational management

AQA A-LEVEL BUSINESS 7132

3.4.1 Operational objectives

- **Operational objectives named by AQA:** costs, quality, speed of response and flexibility, environmental objectives, added value
- **Unit cost objectives:** e.g. cut cost per unit by 5%
- **Quality objectives:** e.g. reduce the defect rate or complaints
- **Speed of response and flexibility:** how quickly the business can react to orders and changes in demand
- **Environmental objectives:** e.g. cut waste, energy use or emissions
- **Added value:** selling price − cost of bought-in materials
- **Influences:** corporate objectives, the market, resources, competitors

3.4.2 Operations data

- **Labour productivity:** output ÷ number of employees (per period)
- **Unit cost (average cost):** total costs ÷ units of output
- **Capacity:** the maximum output a business can produce in a period
- **Capacity utilisation:** actual output ÷ maximum capacity × 100
- **Trap:** capacity utilisation and labour productivity measure different things
- **Using the data:** compare over time and with rivals before judging

3.4.3 Capacity

- **Why capacity matters:** limits how much can be sold; affects unit costs
- **Under-utilisation:** fixed costs spread over fewer units, so unit costs rise
- **Over-utilisation (near 100%):** no room for extra orders, staff and machines stretched, no time for maintenance
- **Ways to raise utilisation:** increase sales, or reduce capacity (rationalisation)
- **Ways to increase capacity:** new equipment, extra shifts, outsourcing
- **Trap:** high utilisation is not always good

3.4.3 Efficiency and productivity

- **Efficiency:** producing output with as few resources as possible
- **Ways to raise productivity:** training, better equipment, better motivation, improved methods
- **Difficulties:** cost of training and equipment, staff resistance, quality may fall if rushed
- **Labour intensive:** relies more on people than machines, e.g. hairdressing
- **Capital intensive:** relies more on machinery and technology than labour, e.g. car assembly
- **Capital intensive:** lower variable cost per unit but high fixed costs and less flexible

3.4.3 Technology in operations

- **Uses of technology:** automation, robotics, design software, stock control systems
- **Benefits:** higher productivity, consistent quality, lower unit costs at high volume
- **Drawbacks:** high initial cost, training, breakdowns, staff resistance, job losses
- **Effect on costs:** raises fixed costs but can lower variable cost per unit
- **Value:** depends on volume of output and how fast the technology goes out of date
- **Trap:** technology only lowers unit costs if output is high enough to spread its cost

3.4.3 Lean production and just in time

- **Lean production:** cutting all forms of waste
- **Types of waste:** excess stock, defects, waiting time, unnecessary movement, overproduction
- **Just in time (JIT):** stock arrives just before it is needed, so little is held
- **Just in case:** holding buffer stock in case of problems
- **JIT benefits:** less cash tied up, less storage space, less waste
- **JIT drawbacks:** depends on reliable suppliers; one late delivery can stop production
- **Kaizen:** continuous small improvements, often suggested by staff
- **Trap:** lean production is not the same as capital intensive

3.4.4 Quality assurance and quality control

- **Quality:** meeting customer expectations consistently
- **Quality control:** inspecting output to find faults, usually at the end
- **Quality assurance:** building quality into every stage to prevent faults
- **Quality assurance relies on:** trained staff who take responsibility for their own work
- **Quality control drawback:** faults found late, so waste and rework
- **Trap:** inspecting finished products is quality control, not quality assurance

3.4.4 Improving quality

- **Benefits of better quality:** fewer returns and complaints, reputation, can charge a higher price, less waste
- **Difficulties:** training costs, time, staff commitment, may slow output at first
- **Consequences of poor quality:** lost customers, rework costs, damaged brand, legal action
- **Customer expectations:** what counts as good quality depends on the price and the market
- **Measuring quality:** e.g. defect rates, returns, complaints, customer satisfaction
- Quality is especially important where a business competes on reliability

3.4.5 Matching supply to demand

- **Ways named by AQA:** outsourcing, temporary and part-time staff, producing to order
- **Outsourcing:** using another business to carry out some work
- **Temporary and part-time staff:** increase or decrease labour to match demand
- **Producing to order:** only make what customers have ordered
- **Benefits:** avoid wasted capacity and unsold stock
- **Drawbacks:** less control, quality risk, may cost more per unit
- **Seasonal demand:** matching supply is especially important

3.4.5 Inventory control

- **Inventory control chart:** shows stock levels over time
- **Lead time:** time between ordering and receiving stock
- **Re-order level:** stock level at which a new order is placed
- **Buffer inventory:** minimum stock kept for emergencies
- **Re-order quantity:** amount ordered each time
- **Maximum stock:** buffer inventory + re-order quantity
- **Holding too much stock:** ties up cash, storage costs, risk of waste
- **Holding too little stock:** risk of running out and losing sales

3.4.5 Suppliers and supply chains

- **Choosing suppliers:** price, quality, reliability, flexibility, payment terms, ethics
- **Supply chain:** the stages from raw materials to the final customer
- **Managing the supply chain:** coordinating suppliers so the right goods arrive at the right time
- **Value of outsourcing:** specialist skills, lower costs, focus on core activities
- **Risks of outsourcing:** less control, quality, dependence on the supplier
- **Dependable supplier:** right quantity and quality, on time, every time